

Experiment 5

Question: Data Collection and Initial Exploration.

Code

```
import json
import re
from collections import Counter

import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import seaborn as sns

def exp_5():
    # Data collection: here we build it; in practice use read_csv / read_excel /
    # read_json / read_sql / web scraping / APIs
    rng = np.random.default_rng(7)
    n = 150
    df = pd.DataFrame({
        "StudentID": range(1, n + 1),
        "Gender": rng.choice(["M", "F"], n),
        "Math": rng.integers(35, 100, n),
        "Science": rng.integers(30, 100, n),
        "Attendance": rng.normal(80, 10, n).round(1),
    })
    df.to_csv("collected_data.csv", index=False)
    df = pd.read_csv("collected_data.csv")

    # Initial exploration
    print("Shape      :", df.shape)
    print("Columns      :", list(df.columns))
    print("Data types   :\n", df.dtypes, "\n")
    print("First 5 rows :\n", df.head(), "\n")
    print("Last 5 rows  :\n", df.tail(), "\n")
    df.info()
    print("\nSummary stats :\n", df.describe(), "\n")
    print("Missing values:\n", df.isnull().sum(), "\n")
    print("Duplicates   :", df.duplicated().sum())
    print("Gender counts :\n", df["Gender"].value_counts())
    print("Correlation   :\n", df[["Math", "Science", "Attendance"]].corr())

    sns.pairplot(df[["Math", "Science", "Attendance", "Gender"]], hue="Gender")
    plt.show()
```

```
exp_5()
```

Output

```
Shape      : (150, 5)
Columns     : ['StudentID', 'Gender', 'Math', 'Science', 'Attendance']
Data types  :
  StudentID      int64
  Gender         str
  Math           int64
  Science        int64
  Attendance     float64
dtype: object
```

```
First 5 rows :
   StudentID Gender  Math  Science  Attendance
0          1     F    89      82         63.6
1          2     F    89      47         85.1
2          3     F    66      85         79.4
3          4     F    96      86         84.1
4          5     F    74      77         70.1
```

```
Last 5 rows :
   StudentID Gender  Math  Science  Attendance
145        146     F    62       70         65.9
146        147     M    45       57         88.5
147        148     F    68       74         87.1
148        149     F    70       48         78.5
```

149 150 M 96 72 62.9

```
<class 'pandas.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
#   Column      Non-Null Count  Dtype
---  -
0   StudentID    150 non-null    int64
1   Gender       150 non-null    str
2   Math         150 non-null    int64
3   Science      150 non-null    int64
4   Attendance   150 non-null    float64
dtypes: float64(1), int64(3), str(1)
memory usage: 6.0 KB
```

```
Summary stats :
      StudentID      Math      Science  Attendance
count  150.000000  150.000000  150.000000  150.000000
mean    75.500000   69.500000   64.653333   79.497333
std     43.445368   19.584184   20.293811    9.620706
min      1.000000   35.000000   30.000000   47.500000
25%     38.250000   53.000000   47.000000   73.775000
50%     75.500000   70.000000   68.000000   79.850000
75%    112.750000   88.500000   82.000000   85.100000
max    150.000000   99.000000   98.000000  100.300000
```

```
Missing values:
StudentID      0
Gender         0
Math           0
Science        0
Attendance     0
dtype: int64
```

```
Duplicates      : 0
Gender counts :
Gender
F      78
M      72
Name: count, dtype: int64
Correlation      :
```

	Math	Science	Attendance
Math	1.000000	-0.164341	-0.034349
Science	-0.164341	1.000000	0.040950
Attendance	-0.034349	0.040950	1.000000

Graph

